

On the Relations between Matchmaking Public Key Encryption and Public Key Authenticated Encryption with Keyword Search

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Abstract

Ateniese et al. (CRYPTO 2019/JoC 2021) introduced a cryptographic primitive which they call matchmaking encryption (ME), and Identity-based ME (IB-ME) is its identity-based variant. IB-ME supports an equality matching where a sender (encryptor) indicates a receiver's (decryptor's) identity (*rcv*) in addition to their own ID (σ), and a receiver indicates a sender's identity (*snd*) in addition to the own identity (ρ). A ciphertext is decrypted if $(\sigma, \rho) = (\text{snd}, \text{rcv})$. In this paper, we pay attention to the search condition of public key authenticated encryption with keyword search (PAEKS) (Huang-Li, Information Sciences 2017) is reminiscent of the equality matching. We introduce a public key variant of ME which we call PK-ME, and propose a generic construction of PK-ME from PAEKS. As a conceptual contribution, our work lies in revealing a connection between ME and public key searchable encryption, which were independently researched so far. Due to the generic construction of PAEKS (Li-Boyen, IACR CiC 2024), we can instantiate the proposed generic construction from pairings or lattices. We also introduce a weaker version of authenticity and show that it can be instantiated from several complexity assumptions. Finally, we discuss the advantage/disadvantage of PK-ME compared to IB-ME.

1 Introduction

1.1 Matchmaking Encryption

In 2019, Ateniese et al. [3, 4] introduced a cryptographic primitive which they call matchmaking encryption (ME) as a non-interactive variant of secret handshake or affiliation-Hiding group key exchange protocols. A sender (encryptor) indicates a policy of a receiver (decryptor), and simultaneously, a receiver also indicates a policy of a sender. A ciphertext is decrypted if both sender attributes satisfy the sender policy and receiver attributes satisfy the receiver policy.

IB-ME, an identity-based variant of ME, is also introduced where the policy is restricted as an equality matching: a sender (encryptor) indicates a receiver's (decryptor's) identity (*rcv*) in addition to the own ID (σ), and a receiver indicates a sender's identity (*snd*) in addition to the own identity (ρ). A ciphertext is decrypted if

$$(\sigma, \rho) = (\text{snd}, \text{rcv})$$

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As a special requirement, when the decryption fails, then no information beyond the decryption failure is revealed, i.e., a receiver does not know whether $\text{rcv} \neq \rho$ or $\text{snd} \neq \sigma$. Ateniese et al. proposed an IB-ME scheme which is secure in the random oracle model. As subsequent works, Francati et al. [24] and Chen et al. [11] proposed IB-ME schemes secure in the standard model. Francati et al. [24] introduced a stronger flavor of privacy which they call enhanced privacy. It captures privacy according to every possible mismatch condition for the receiver where an adversary holds a decryption key but does not know the sender's identities. Chiku et al. [14] proposed an efficient IB-ME scheme in the random oracle model. Wang et al. [41] proposed a generic construction of IB-ME from 2-level anonymous hierarchical identity-based encryption (HIBE) and identity-based signature (IBS) and gave a lattice-based instantiation. Boyen and Li [7] also proposed a generic construction of IB-ME from IBE, an average-case randomness extractor, and a reusable computational extractor, and gave several instantiations. Lin et al. [32] considers a chosen-ciphertext security. Chiku et al. [13] proposed a key-escrow free IB-ME scheme by considering IBE with security against the key generation center (KGC) [15, 22, 23]. As another attempt to solve the key-escrow problem, Chen et al. [10] and Yang et al. [42] proposed certificateless ME, and Francati et al. [25] proposed registered ME. Chu et al. [16] proposed inner-product ME that supports more expressive policy than that of IB-ME. Jiang et al. [30] proposed revocable IB-ME. As mentioned above, ME has been widely researched from the viewpoint of several aspects.

1.2 Public Key Authenticated Encryption with Keyword Search

As a searchable encryption in a public key setting, public key encryption with keyword search (PEKS) [5] has been proposed. A keyword kw is encrypted using the receiver's public key and a trapdoor associated to a keyword kw' is generated using the receiver's secret key. The test algorithm that takes a ciphertext and a trapdoor outputs 1 if $kw = kw'$ and 0, otherwise. Abdalla et al. [1] proposed a generic construction of PEKS from anonymous IBE.

It is well-known that the following keyword guessing attack works in PEKS where an adversary, who has a trapdoor, arbitrarily chooses a keyword kw , generates its ciphertext, and checks whether the trapdoor is associated to kw or not by running the test algorithm. The keyword guessing attack is inevitable in PEKS since anyone can freely produce a ciphertext of a keyword using the receiver's public key. To prevent the keyword guessing attack, Huang and Li [29] introduced public key authenticated encryption with keyword search (PAEKS) that provides TI (trapdoor indistinguishability) where no information of keyword is revealed from trapdoors. In PAEKS, a keyword kw is encrypted by using a secret key of a sender S and a public key of a receiver R . To generate a trapdoor for a keyword kw' , a public key of a sender S' and a secret key of a receiver R' are used. The test algorithm outputs 1 when

$$(S, R, kw) = (S', R', kw')$$

Moreover, consistency for multi users was introduced in [21] where the test algorithm outputs 0 when $(S, R, kw) \neq (S', R', kw')$. Note that it is not required to reveal information whether $S \neq S'$, $R \neq R'$, or $kw \neq kw'$ when the test algorithm outputs 0.

Li and Boyen [31] proposed a generic construction of PAEKS from symmetric-key equality-predicate encryption (EPE) (which they introduced) and non-interactive key exchange (NIKE) [26]. They instantiated EPE by pseudo-random function (PRF), pairings, or lattices. The generic construction can be instantiated by several complexity assumptions such as pairing-free computational/decisional Diffie-Hellman, pairings, integer factorizations, lattices, and isogenies based on several NIKes [8, 9, 18, 27]. Later, Shinoki et al. [39] pointed out a flaw of Li-Boyen's pairing-based

EPE construction, and proposed another pairing-based EPE construction.¹

For TI, there are two security notions: no information of keyword is revealed from trapdoors when an adversary is (resp. is not) allowed to send challenge keywords to the trapdoor oracle. The former is called full TI and the latter is called TI. Full TI is stronger than TI, and the Li-Boyer construction [31] instantiates PAEKS schemes with the full TI security from pairings or lattices.

Our Motivation. We pay attention to the search condition of PAEKS, $(S, R, kw) \stackrel{?}{=} (S', R', kw')$ as above, is reminiscent of the equality matching of IB-ME, $(\sigma, \rho) \stackrel{?}{=} (\text{snd}, \text{rcv})$ as above. As a natural observation, there is a relation between ME and PAEKS. However, no explicit implication result has been mentioned between them so far, to the best of our knowledge. Thus, as a natural question:

Is there any relation between ME and searchable encryption?

1.3 Our Contribution

In this paper, we explicitly show the relation between ME and searchable encryption. Concretely,

1. We introduce a public key variant of ME which we call PK-ME.²
 - We define security notions of PK-ME, privacy and authenticity, by considering the security notions of IB-ME.
 - We also discuss the merit/demerit of PK-ME compared to IB-ME. Especially, we explore whether PK-ME can be used for realizing applications of IB-ME considered by Ateniese et al. [3, 4].
2. We propose a generic construction of PK-ME from PAEKS.
 - As a conceptual contribution, our work lies in revealing a connection between ME and public key searchable encryption, which were basically researched independently.

Technical Overview. Basically, we follow the generic construction of IBE from PEKS given by Boneh et al. [5]. Briefly, a plaintext $M \in \{0, 1\}$ is regarded as a keyword and is encrypted by the underlying PAEKS scheme. The decryption key consists of two trapdoors for two keywords 0 and 1. The PK-ME decryption algorithm outputs M if the test algorithm with the trapdoor of M outputs 1 where $M \in \{0, 1\}$.

Here, we point out that the Boneh et al.'s generic construction did not consider the case where the test algorithm with the trapdoor of 0 outputs 1, and with the trapdoor of 1 also outputs 1. Then, the decryption result is not uniquely determined and correctness does not hold. To provide correctness, (1) we additionally introduce **Uncertain** as an output of the decryption algorithm, and (2) prove that the decryption algorithm does not output **Uncertain** if the underlying PAEKS scheme provides consistency. In addition to this, we introduce key privacy of PAEKS, that is not usually required as a security of PAEKS, to hide information of sender and receiver from ciphertexts. Later, we explain the Li-Boyer generic construction of PAEKS provides key privacy.

¹Shinoki et al. [39] called EPE SSE (searchable symmetric encryption).

²MPKE or MMPKE might be straightforward when we directly abbreviate matchmaking public key encryption. However, Alwen et al. [2] have introduced them as abbreviations for multi-recipient public-key encryption (mpKE) and multi-message multi-recipient PKE (mmPKE), respectively. By considering IB-ME, we named PK-ME as the abbreviation of matchmaking public key encryption.

Since (PK-)ME does not consider information leakage from decryption keys, intuitively, the TI security of PAEKS is not necessary to construct PK-ME. Counterintuitively, our security proof (of authenticity) is reduced to full TI which seems to be the most non-trivial part of our security proof. Thus, our construction requires that the underlying PAEKS is to be fully TI secure, which can be instantiated from pairings or lattices via the Li-Boyen generic construction. We also introduce a weaker version of authenticity that does not require the full TI security of the underlying PAEKS scheme. Because it just requires the TI security of the underlying PAEKS scheme, we can instantiate our generic construction from (pairing-free) computational Diffie-Hellman, integer factorization, or isogenies via the Li-Boyen generic construction with NIKes [8, 9, 18, 27]. Weak authenticity seems somewhat artificial, and it is not clear what the real situation is captured under the security model. Nevertheless, we claim that introducing such an authenticity is meaningful to clarify what the security level can be realized by only the TI security.

Remark. If PAEKS can be constructed from PK-ME, then two cryptographic primitives are equivalent. However, we expect that PK-ME does not imply PAEKS because PK-ME does not consider information leakage from decryption keys whereas PAEKS requires that no information of keyword is leaked from trapdoors. Moreover, key privacy is not always required in PAEKS. Exploring this expectation is an interesting open problem.

1.4 Related Work

Inspired by ME, Mongardín et al. [35] introduced identity-based matchmaking signatures (IB-MSS) and proposed a (semi-)generic construction³ of IB-MSS from identity-based non-interactive key distribution (IB-NIKD) [37] and signatures. IB-NIKD can be seen as an identity-based variant of NIKE, and NIKE is a building block of the Li-Boyen generic construction of PAEKS as mentioned above. We observe that these relations between non-interactive KD/KE and ME variants may suggest that our PK-ME construction from PAEKS is reasonable and natural.

Chiku et al. [14] mentioned the difference/similarity between IB-ME and identity-based signcryption [34] because both primitives consider privacy and authenticity simultaneously. They claimed that IB-ME provides better security properties than identity-based signcryption because IB-ME ensures anonymity of users and the confidentiality of messages when ciphertexts are tried to be decrypted with mismatched sender identities. Chiku et al. also proposed a generic construction of IB-ME from IBE, identity-based signatures, and reusable extractors.

Wang et al. [40] proposed a searchable asymmetric encryption (SAE) with bilateral access control. They mentioned that their security model is inspired by ME. Though this attempt connects ME and searchable encryption, they did not mention a similarity of the decryption condition of IB-ME and the search condition of PAEKS, and no explicit implication result was given. To sum up, no explicit implication result has been mentioned between ME and searchable encryption so far, to the best of our knowledge.

Hovd and Stam proposed vetted encryption (VE) [28], and discussed differences/similarities between VE and IB-ME. Especially, they mentioned that VE does not require the KGC unlike IB-ME. Since one difference between PK-ME and IB-ME is whether the KGC is required or not, we discuss differences between PK-ME and VE here, for the sake of clarity. VE introduces an outside filter that a recipient restricts who can send a message to the recipient. There are three variants by whether a sender is identified or not. In anonymous vetted encryption (AVE), both the filter and the recipient do not identify a sender. The authors mentioned that AVE is reminiscent of

³Mongardín et al. assumed that the form of a public key is represented as g^{sk} , and sk is generated via IB-KD in their IB-MSS construction.

group signatures in this perspective. In identifiable vetted encryption (IVE), both the filter and the recipient identify a sender. The authors mentioned that AVE is reminiscent of signcryption in this perspective. In opaque vetted encryption (OVE), the filter does not identify a sender whereas the recipient identifies the sender. The authors mentioned that AVE is reminiscent of group signatures and its opening functionality in this perspective. A difference between OVE and IB-ME besides the KGC is how to indicate a sender. The decryption algorithm takes a sender identity snd in IB-ME whereas the decryption result contains a sender identity in OVE. Moreover, a sender indicates a receiver identity and a receiver indicates a sender identity, respectively, in IB-ME. That is, a sender and a receiver indicate the partner with each other. In VE, only a recipient specifies a sender. This difference is the same as that of between PK-ME and VE.

Differences from the proceedings version. An extended abstract appeared at the 24th International Conference on Cryptology and Network Security (CANS) 2025 [43]. In this full version, we introduce a weaker version of authenticity that can be achieved by only the TI security, and demonstrate that our generic construction with weak authenticity can be instantiated from several complexity assumptions such as pairing-free computational/decisional Diffie-Hellman, pairings, integer factorizations, lattices, and isogenies.

Concurrent and Independent Work. Ling et al. [33] also proposed a pairing-based PK-ME scheme which is secure in the random oracle model. One of the biggest differences from us is that they have not discovered the relationship between searchable encryption and PK-ME. Although they also cite the PAEKS paper [31], they state that “*these are far from achieving our goal*” (Section 1.2. Related work in [33]) and are not actually related to the construction. Our primary objective is to investigate a non-trivial relationship between searchable encryption and matchmaking encryption, and our generic construction of PK-ME from PAEKS serves as a means to illustrate this connection. From this perspective, we view their results as concurrent and independent work that does not lessen the value of our contribution.

2 Preliminaries: Definitions of PAEKS

We define a PAEKS scheme $\text{PAEKS} = (\text{PAEKS.SetUp}, \text{PAEKS.KGen}_S, \text{PAEKS.KGen}_R, \text{PAEKS.Enc}, \text{Trapdoor}, \text{Test})$ as follows.

Definition 1 (Syntax of PAEKS).

$\text{PAEKS.SetUp}(1^\lambda)$: The setup algorithm takes a security parameter $\lambda \in \mathbb{N}$ as input, and outputs a common parameter pp_{PAEKS} . Assume that pp_{PAEKS} contains a keyword space $\mathcal{KW}$.

$\text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$: The sender key generation algorithm takes pp_{PAEKS} as input, and outputs a sender’s public key PAEKS.pk_S and a sender’s secret key PAEKS.sk_S .

$\text{PAEKS.KGen}_R(\text{pp}_{\text{PAEKS}})$: The receiver key generation algorithm takes pp_{PAEKS} as input, and outputs a receiver’s public key PAEKS.pk_R and a receiver’s secret key PAEKS.sk_R .

$\text{PAEKS.Enc}(\text{PAEKS.pk}_S, \text{PAEKS.pk}_R, \text{PAEKS.sk}_S, kw)$: The keyword encryption algorithm takes PAEKS.pk_S , PAEKS.pk_R , PAEKS.sk_S , and a keyword $kw \in \mathcal{KW}$ as input, and outputs a ciphertext $\text{PAEKS.ct}_{S,R,kw}$.

$\text{Trapdoor}(\text{PAEKS.pk}_S, \text{PAEKS.pk}_R, \text{PAEKS.sk}_R, kw)$: The trapdoor generation algorithm takes PAEKS.pk_S , PAEKS.pk_R , PAEKS.sk_R , and $kw \in \mathcal{KW}$ as input, and outputs a trapdoor $\text{td}_{S,R,kw}$.

$\text{Test}(\text{PAEKS.ct}_{S,R,kw}, \text{td}_{S',R',kw'})$: The test algorithm takes $\text{PAEKS.ct}_{S,R,kw}$ and $\text{td}_{S',R',kw'}$ as input, and outputs 1 or 0.

Next, we define correctness. It guarantees that for an honestly generated ciphertext $\text{PAEKS.ct}_{S,R,kw}$ and honestly generated trapdoor $\text{td}_{S',R',kw'}$, the Test algorithm outputs 1 if $(S, R, kw) = (S', R', kw')$.

Definition 2 (Correctness). For all $\lambda \in \mathbb{N}$, $\text{pp}_{\text{PAEKS}} \leftarrow \text{PAEKS.Setup}(1^\lambda)$, $(\text{PAEKS.pk}_S, \text{PAEKS.sk}_S) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$, $(\text{PAEKS.pk}_R, \text{PAEKS.sk}_R) \leftarrow \text{PAEKS.KGen}_R(\text{pp}_{\text{PAEKS}})$, and $kw \in \mathcal{KW}$, we say that PAEKS is correct if $\text{Test}(\text{PAEKS.Enc}(\text{PAEKS.pk}_S, \text{PAEKS.pk}_R, \text{PAEKS.sk}_S, kw), \text{Trapdoor}(\text{PAEKS.pk}_S, \text{PAEKS.pk}_R, \text{PAEKS.sk}_R, kw)) = 1$.

Next, we define consistency. It guarantees that for an honestly generated ciphertext $\text{PAEKS.ct}_{S,R,kw}$ and honestly generated trapdoor $\text{td}_{S',R',kw'}$, the Test algorithm outputs 0 if $(S, R, kw) \neq (S', R', kw')$. Let $\mathcal{A}$ be a probabilistic polynomial-time (PPT) adversary and $\mathcal{C}$ be the challenger.

Definition 3 (Consistency). $\mathcal{C}$ runs

- $\text{pp}_{\text{PAEKS}} \leftarrow \text{PAEKS.Setup}(1^\lambda)$
- $(\text{PAEKS.pk}_{S_0}, \text{PAEKS.sk}_{S_0}) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$
- $(\text{PAEKS.pk}_{S_1}, \text{PAEKS.sk}_{S_1}) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$
- $(\text{PAEKS.pk}_{R_0}, \text{PAEKS.sk}_{R_0}) \leftarrow \text{PAEKS.KGen}_R(\text{pp}_{\text{PAEKS}})$
- $(\text{PAEKS.pk}_{R_1}, \text{PAEKS.sk}_{R_1}) \leftarrow \text{PAEKS.KGen}_R(\text{pp}_{\text{PAEKS}})$

and gives $(\text{pp}_{\text{PAEKS}}, \text{PAEKS.pk}_{S_0}, \text{PAEKS.pk}_{S_1}, \text{PAEKS.pk}_{R_0}, \text{PAEKS.pk}_{R_1})$ to $\mathcal{A}$. $\mathcal{A}$ outputs $(i, j, kw, i', j', kw') \in (\{0, 1\}^2 \times \mathcal{KW})^2$. $\mathcal{C}$ computes $\text{PAEKS.ct}_{S_i, R_j, kw} \leftarrow \text{PAEKS.Enc}(\text{PAEKS.pk}_{S_i}, \text{PAEKS.pk}_{R_j}, \text{PAEKS.sk}_{S_i}, kw)$ and $\text{td}_{S_{i'}, R_{j'}, kw'} \leftarrow \text{Trapdoor}(\text{PAEKS.pk}_{S_{i'}}, \text{PAEKS.pk}_{R_{j'}}, \text{PAEKS.sk}_{R_{j'}}, kw')$. $\mathcal{A}$ wins if $\text{Test}(\text{PAEKS.ct}_{S_i, R_j, kw}, \text{td}_{S_{i'}, R_{j'}, kw'}) = 1$ and $(S_i, R_j, kw) \neq (S_{i'}, R_{j'}, kw')$ holds. We say that PAEKS is consistent if for any PPT adversaries $\mathcal{A}$, the advantage $\text{Adv}_{\text{PAEKS}, \mathcal{A}}^{\text{consist}}(\lambda) = \Pr[\mathcal{A} \text{ wins}]$ is negligible in λ .

Next, we define full ciphertext-keyword indistinguishability (full CI).⁴ It guarantees that no information of keyword is revealed from ciphertexts. Here, full means that an adversary is allowed to obtain ciphertexts of the challenge keywords. We also consider key privacy where a ciphertext does not leak information of the sender $(\text{PAEKS.pk}_S, \text{PAEKS.sk}_S)$ and the receiver (PAEKS.pk_R) . Let $\mathcal{C}$ be the challenger and $\mathcal{A}$ be an adversary.

Definition 4 (Full CI with Key Privacy). $\mathcal{C}$ runs $\text{pp}_{\text{PAEKS}} \leftarrow \text{PAEKS.Setup}(1^\lambda)$, $(\text{PAEKS.pk}_{S_0}, \text{PAEKS.sk}_{S_0}) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$, $(\text{PAEKS.pk}_{S_1}, \text{PAEKS.sk}_{S_1}) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$, $(\text{PAEKS.pk}_{R_0}, \text{PAEKS.sk}_{R_0}) \leftarrow \text{PAEKS.KGen}_R(\text{pp}_{\text{PAEKS}})$, and $(\text{PAEKS.pk}_{R_1}, \text{PAEKS.sk}_{R_1}) \leftarrow \text{PAEKS.KGen}_R(\text{pp}_{\text{PAEKS}})$, and gives $(\text{pp}_{\text{PAEKS}}, \text{pk}_{S_0}, \text{pk}_{S_1}, \text{pk}_{R_0}, \text{pk}_{R_1})$ to $\mathcal{A}$.

$\mathcal{A}$ declares the challenge keywords $(kw_0^*, kw_1^*) \in \mathcal{KW}^2$. $\mathcal{C}$ chooses $b \xleftarrow{\$} \{0, 1\}$, computes the challenge ciphertext $\text{PAEKS.ct}_{S_b, R_b, kw_b^*}^* \leftarrow \text{PAEKS.Enc}(\text{PAEKS.pk}_{S_b}, \text{PAEKS.pk}_{R_b}, \text{PAEKS.sk}_{S_b}, kw_b^*)$, and returns $\text{PAEKS.ct}_{S_b, R_b, kw_b^*}^*$ to $\mathcal{A}$.

During the game, $\mathcal{A}$ is allowed to issue the following queries.

⁴In the definition by Li-Boyen, an adversary is allowed to issue encryption/trapdoor queries by using a public key that is chosen by the adversary. In this paper, our main aim is to provide a minimum functionality as matchmaking encryption, and thus we do not consider such an adversarial public key.

Encryption query : $\mathcal{A}$ sends $(i, j, kw) \in \{0, 1\}^2 \times \mathcal{KW}$ to $\mathcal{C}$. $\mathcal{C}$ computes $\text{PAEKS.ct}_{S_i, R_j, kw} \leftarrow \text{PAEKS.Enc}(\text{PAEKS.pk}_{S_i}, \text{PAEKS.pk}_{R_j}, \text{PAEKS.sk}_{S_i}, kw)$, and returns $\text{PAEKS.ct}_{S_i, R_j, kw}$ to $\mathcal{A}$. There is no restriction here.

Trapdoor query : $\mathcal{A}$ sends $(i, j, kw) \in \{0, 1\}^2 \times \mathcal{KW}$ to $\mathcal{C}$. Here, $\mathcal{A}$ is not allowed to send $(0, 0, kw_0^*)$ or $(1, 1, kw_1^*)$. $\mathcal{C}$ computes $\text{td}_{S_i, R_j, kw} \leftarrow \text{Trapdoor}(\text{PAEKS.pk}_{S_i}, \text{PAEKS.pk}_{R_j}, \text{PAEKS.sk}_{R_j}, kw)$, and returns $\text{td}_{S_i, R_j, kw}$ to $\mathcal{A}$.

Finally, $\mathcal{A}$ outputs $b' \in \{0, 1\}$. We say that PAEKS provides the full CI security if for all PPT adversaries $\mathcal{A}$, the advantage

$$\text{Adv}_{\text{PAEKS}, \mathcal{A}}^{\text{Full-KP-CI}}(\lambda) = |\Pr[b = b'] - 1/2|$$

is negligible in λ .

Note that key privacy does not always hold. For example, the Chen et al.'s pairing-based full CI secure PAEKS scheme [12] does not provide key privacy because for $\text{pk}_R = g^x$, a ciphertext contains g^{xr} and g^r that reveal information of pk_R (via $e(\text{pk}_R, g^r) = e(g^{xr}, g)$ holds or not). We emphasize that this is not an attack against the Chen et al.'s PAEKS scheme. On the other hand, the generic construction of Li-Boyen [31] provides key privacy. In the generic construction, a NIKE key, defined by public keys of a sender S and a receiver R , is used as an encryption key of EPE. Due to the security of NIKE, the NIKE key is indistinguishable from random. Thus, a PAEKS ciphertext does not leak information of S and R . We employed PAEKS schemes instantiated by the Li-Boyen generic construction later.

Next, we define full trapdoor indistinguishability (full TI). It guarantees that no information of keyword is revealed from trapdoors. Here, full means that an adversary is allowed to obtain trapdoors of the challenge keywords.

Definition 5 (Full TI). $\mathcal{C}$ runs $\text{pp}_{\text{PAEKS}} \leftarrow \text{PAEKS.Setup}(1^\lambda)$, $(\text{PAEKS.pk}_S, \text{PAEKS.sk}_S) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$, and $(\text{PAEKS.pk}_R, \text{PAEKS.sk}_R) \leftarrow \text{PAEKS.KGen}_R(\text{pp}_{\text{PAEKS}})$, and gives $(\text{pp}_{\text{PAEKS}}, \text{PAEKS.pk}_S, \text{PAEKS.pk}_R)$ to $\mathcal{A}$.

$\mathcal{A}$ declares the challenge keywords $(kw_0^*, kw_1^*) \in \mathcal{KW}^2$. $\mathcal{C}$ chooses $b \xleftarrow{\$} \{0, 1\}$, computes the challenge trapdoor $\text{td}_{S, R, kw_b^*} \leftarrow \text{Trapdoor}(\text{PAEKS.pk}_S, \text{PAEKS.pk}_R, \text{PAEKS.sk}_R, kw_b^*)$, and returns td_{S, R, kw_b^*} to $\mathcal{A}$.

During the game, $\mathcal{A}$ is allowed to issue the following queries.

Encryption query : $\mathcal{A}$ sends kw to $\mathcal{C}$. Here, $\mathcal{A}$ is not allowed to send kw_0^* or kw_1^* . $\mathcal{C}$ computes $\text{PAEKS.ct}_{S, R, kw} \leftarrow \text{Enc}(\text{PAEKS.pk}_S, \text{PAEKS.pk}_R, \text{PAEKS.sk}_S, kw)$ and returns $\text{PAEKS.ct}_{S, R, kw}$ to $\mathcal{A}$.

Trapdoor query : $\mathcal{A}$ sends kw to $\mathcal{C}$. $\mathcal{C}$ computes $\text{td}_{S, R, kw} \leftarrow \text{Trapdoor}(\text{PAEKS.pk}_S, \text{PAEKS.pk}_R, \text{PAEKS.sk}_R, kw)$ and returns $\text{td}_{S, R, kw}$ to $\mathcal{A}$. There is no restriction here.

Finally, $\mathcal{A}$ outputs $b' \in \{0, 1\}$. We say that PAEKS provides the full TI security if for all PPT adversaries $\mathcal{A}$, the advantage

$$\text{Adv}_{\text{PAEKS}, \mathcal{A}}^{\text{Full-TI}}(\lambda) = |\Pr[b = b'] - 1/2|$$

is negligible in λ .

If the adversary is not allowed to send the challenge keywords kw_0^* and kw_1^* as a trapdoor query, then the security definition is said to be TI.

3 Definitions of PK-ME

3.1 Syntax of PK-ME

In this section, we newly define PK-ME $\text{PK-ME} = (\text{SetUp}, \text{SKGen}, \text{RKGen}, \text{Enc}, \text{Dec})$ as follows. To clarify the relations between PK-ME and IB-ME, we introduce an index space $\mathcal{IDX}$ (which is not an identity space) and $\sigma, \rho \in \mathcal{IDX}$ are just indexes unlike IB-ME. Then, similar to IB-ME, we can describe the decryption condition of PK-ME: a ciphertext generated by $(\text{ek}_\sigma, \text{pk}_{\text{rcv}})$ can be decrypted by $(\text{dk}_\rho, \text{pk}_{\text{snd}})$ if $(\sigma, \rho) = (\text{snd}, \text{rcv})$. We note that, in addition to a plaintext M and a failure symbol $\perp$, the decryption algorithm Dec may output **Uncertain** which means that the decryption result is not uniquely determined. We will explain the reason behind later.

Definition 6 (Syntax of PK-ME).

$\text{SetUp}(1^\lambda)$: The setup algorithm takes a security parameter $\lambda \in \mathbb{N}$ as input, and outputs a common parameter pp . Assume that pp contains a plaintext space $\mathcal{MSG}$ and an index space $\mathcal{IDX}$.

$\text{SKGen}(\text{pp}, \sigma)$: The sender key generation algorithm takes pp and index $\sigma \in \mathcal{IDX}$ as input, and outputs a sender's public key pk_σ and sender's encryption key ek_σ .

$\text{RKGen}(\text{pp}, \rho)$: The receiver key generation algorithm takes pp and index $\rho \in \mathcal{IDX}$ as input, and outputs a receiver's public key pk_ρ and receiver's decryption key dk_ρ .

$\text{Enc}(\text{pp}, \text{ek}_\sigma, \text{pk}_{\text{rcv}}, M)$: The encryption algorithm takes pp , ek_σ , an intended receiver's public key pk_{rcv} , where $\text{rcv} \in \mathcal{IDX}$, and a plaintext M as input, and outputs a ciphertext ct .

$\text{Dec}(\text{pp}, \text{dk}_\rho, \text{pk}_{\text{snd}}, \text{ct})$: The decryption algorithm takes pp , dk_ρ , an intended sender's public key pk_{snd} , where $\text{rcv} \in \mathcal{IDX}$, and ct as input, and outputs M or $\perp$ or **Uncertain**.

3.2 Security Definition

Next, we define correctness. Intuitively, it guarantees that ciphertext generated by $(\text{ek}_\sigma, \text{pk}_{\text{rcv}})$ can be decrypted by $(\text{dk}_\rho, \text{pk}_{\text{snd}})$ if $(\sigma, \rho) = (\text{snd}, \text{rcv})$. Here, we define it in two steps because the output of the decryption algorithm contains **Uncertain**. First (correctness 1), we define the case that the decryption algorithm with honestly generated keys and an honestly generated ciphertext outputs either a plaintext or **Uncertain** with overwhelming probability. Second (correctness 2), we define the case that no PPT adversary can produce a plaintext where the decryption result of the corresponding ciphertext is **Uncertain**. We say that PK-ME is correct if PK-ME provides both correctness 1 and 2.

Definition 7 (Correctness 1). For any $\lambda \in \mathbb{N}$, $\text{pp} \leftarrow \text{SetUp}(1^\lambda)$, $\sigma, \rho \in \mathcal{IDX}$, $(\text{pk}_\sigma, \text{ek}_\sigma) \leftarrow \text{SKGen}(\text{pp}, \sigma)$, $(\text{pk}_\rho, \text{sk}_\rho) \leftarrow \text{RKGen}(\text{pp}, \rho)$, and $M \in \mathcal{MSG}$, we say that PK-ME provides correctness 1 if $\text{Dec}(\text{pp}, \text{dk}_\rho, \text{pk}_\sigma, \text{Enc}(\text{pp}, \text{ek}_\sigma, \text{pk}_\rho, M)) = M/\text{Uncertain}$ holds with overwhelming probability.

Definition 8 (Correctness 2). Let $\mathcal{C}$ be the challenger and $\mathcal{A}$ be an adversary. $\mathcal{C}$ chooses distinct indexes $\sigma, \rho \in \mathcal{IDX}$, computes $\text{pp} \leftarrow \text{SetUp}(1^\lambda)$, $(\text{pk}_\sigma, \text{ek}_\sigma) \leftarrow \text{SKGen}(\text{pp}, \sigma)$, and $(\text{pk}_\rho, \text{sk}_\rho) \leftarrow \text{RKGen}(\text{pp}, \rho)$, and gives $(\text{pp}, \text{pk}_\sigma, \text{pk}_\rho)$ to $\mathcal{A}$. $\mathcal{A}$ outputs $M \in \mathcal{MSG}$. $\mathcal{C}$ computes $\text{ct} \leftarrow \text{Enc}(\text{pp}, \text{ek}_\sigma, \text{pk}_\rho, M)$. $\mathcal{A}$ wins if $\text{Dec}(\text{pp}, \text{dk}_\rho, \text{pk}_\sigma, \text{ct}) = \text{Uncertain}$. We say that PK-ME provides correctness 2 if the advantage

$$\text{Adv}_{\text{PK-ME}, \mathcal{A}}^{\text{correct2}}(\lambda) = \Pr[\mathcal{A} \text{ wins}]$$

is negligible in the security parameter for all PPT adversaries $\mathcal{A}$.

Ateniese et al. [3, 4] defined privacy and authenticity as security notions of IB-ME. We define privacy and authenticity of PK-ME by considering Ateniese et al.'s definitions. For privacy, Ateniese et al. defined that no information of plaintext and key is revealed from ciphertexts for a malicious decryptor who does not have dk_ρ . Here, an adversary is allowed to obtain encryption keys without any restriction. That is, privacy guarantees that no information of which encryption key was used for generating a ciphertext even if the adversary has the encryption key. In the following, we weaken the security level by defining the encryption oracle that reflects the situation where an adversary is allowed to access the encryption oracle but is not allowed to obtain sender's secret keys in the full CI security model. The reason is explained more clearly below. We can enhance the definition where an adversary obtains encryption keys ek_{σ_0} and ek_{σ_1} . However, then we cannot employ a PAEKS scheme instantiated by the (NIKE-based) Li-Boyen generic construction as a building block of the proposed generic construction. A NIKE key can be generated by either (1) the sender's public key and the receiver's secret key or (2) the sender's secret key and the receiver's public key. An encryption key in the proposed generic construction of PK-ME is a sender's secret key in PAEKS. If the challenge sender's key is revealed in the full CI security, the adversary can generate a NIKE key for the challenge keyword and can generate a trapdoor for the challenge keyword that breaks the full CI security. We note that the situation that a sender also can generate a trapdoor does not affect security of PAEKS. We leave improving the security level for future work. Let $\mathcal{C}$ be the challenger and $\mathcal{A}$ be an adversary.

Definition 9 (Privacy). $\mathcal{C}$ computes $pp \leftarrow \text{Setup}(1^\lambda)$, $(pk_{\sigma_0}, ek_{\sigma_0}) \leftarrow \text{SKGen}(pp, \sigma_0)$, $(pk_{\sigma_1}, ek_{\sigma_1}) \leftarrow \text{SKGen}(pp, \sigma_1)$, $(pk_{\rho_0}, dk_{\rho_0}) \leftarrow \text{RKGen}(pp, \rho_0)$, and $(pk_{\rho_1}, dk_{\rho_1}) \leftarrow \text{RKGen}(pp, \rho_1)$, and gives $(pp, pk_{\sigma_0}, pk_{\sigma_1}, pk_{\rho_0}, pk_{\rho_1})$ to $\mathcal{A}$. $\mathcal{A}$ declares (M_0^*, M_1^*) . $\mathcal{C}$ chooses $b \xleftarrow{\$} \{0, 1\}$, computes the challenge ciphertext $ct^* \leftarrow \text{Enc}(pp, ek_{\sigma_b}, pk_{\rho_b}, M_b^*)$, and gives ct^* to $\mathcal{A}$. During the game, $\mathcal{A}$ is allowed to issue the following queries.

Encryption query : $\mathcal{A}$ sends (i, j, M) ($i, j \in \{0, 1\}$) to $\mathcal{C}$. $\mathcal{C}$ computes $ct \leftarrow \text{Enc}(pp, ek_{\sigma_i}, pk_{\rho_j}, M)$, and gives ct to $\mathcal{A}$.

$\mathcal{A}$ outputs $b' \in \{0, 1\}$. We say that PK-ME provides privacy if for all PPT adversaries $\mathcal{A}$, the advantage

$$\text{Adv}_{\text{PK-ME}, \mathcal{A}}^{\text{priv}}(\lambda) = |\Pr[b = b'] - 1/2|$$

is negligible in the security parameter λ .

We remark that, in IB-ME, the adversary of privacy is allowed to access two oracles, $O_1 := \text{SKGen}(\text{msk}, \cdot)$ and $O_2 := \text{SKGen}(\text{msk}, \cdot)$ where msk is the master secret key. The adversary sends an identity σ to O_1 and can obtain ek_σ , and sends an identity ρ to O_2 and can obtain dk_ρ . No pk is defined since each identity works as a public key in IB-ME. We clarify that PK-ME also considers multiple encryption and decryption keys where the number of keys is a polynomial of the security parameter, since the adversary can run the SKGen algorithm and the RKGen algorithm without any oracle access.

For authenticity, Ateniese et al. defined that no adversary who does not have ek_{snd} can produce a ciphertext where the ciphertext is decryptable (i.e., the decryption algorithm does not output $\perp$) when snd is indicated as the sender. We also consider the case that the decryption result is not Uncertain as a winning condition. Let $\mathcal{C}$ be the challenger and $\mathcal{A}$ be an adversary.

Definition 10 (Authenticity). $\mathcal{C}$ computes $pp \leftarrow \text{Setup}(1^\lambda)$, $(pk_\sigma, ek_\sigma) \leftarrow \text{SKGen}(pp, \sigma)$, and $(pk_\rho, dk_\rho) \leftarrow \text{RKGen}(pp, \rho)$, and gives (pp, pk_σ, pk_ρ) to $\mathcal{A}$. $\mathcal{A}$ outputs ct^* . $\mathcal{C}$ computes $M^* \leftarrow$

$\text{Dec}(\text{pp}, \text{dk}_\rho, \text{pk}_\sigma, \text{ct}^*)$. We say that PK-ME provides authenticity if for all PPT adversaries $\mathcal{A}$, the advantage

$$\text{Adv}_{\text{PK-ME}, \mathcal{A}}^{\text{Auth}}(\lambda) = \Pr[(M^* \neq \perp) \cap (M^* \neq \text{Uncertain})]$$

is negligible in the security parameter λ .

As in privacy, the adversary of authenticity in IB-ME is allowed to access two oracles, $\text{O}_1 := \text{SKGen}(\text{msk}, \cdot)$ and $\text{O}_2 := \text{SKGen}(\text{msk}, \cdot)$. As a restriction, the adversary is not allowed to obtain the encryption key for the challenge sender's identity, and is not allowed to obtain the decryption key for the challenge receiver's decryption key. In PK-ME, the adversary can run the SKGen algorithm and the RKGGen algorithm without any oracle access but does not obtain $(\text{ek}_\sigma, \text{dk}_\rho)$. Thus, the definition of authenticity in PK-ME captures the same scenario as in IB-ME, except for the identity-based setting.

Next, we introduce a weaker version of authenticity which we call weak authenticity. Before giving the definition, we discuss why the full TI security is required to prove the authenticity defined in Def. 10 (See the proof of Theorem 3 for detail). In the security proof, the trapdoor of the challenge keyword is used for decrypting the forged ciphertext ct^* . In other words, no such a trapdoor query is required if the reduction algorithm can know the decryption result of ct^* . From this perspective, we can define authenticity w.r.t. message where the challenger indicates the challenge plaintext M^* and an adversary is required to produce a decryptable ciphertext ct^* (without knowing the corresponding encryption key) whose decryption result is M^* . Then, the reduction algorithm does not have to issue a trapdoor query for the challenge keyword. However, this authenticity w.r.t. message is quite weak since an adversary, who does not have the encryption key, may be able to produce a decryptable ciphertext whose decryption result is a non- $\perp$ plaintext $M \neq M^*$. Thus, we define weak authenticity where the adversary declares M^* that is the decryption result of the forged ciphertext ct^* . The security model seems artificial, and it is not clear what the real situation is captured under the security model. Moreover, an adversary may be able to produce a decryptable ciphertext without knowing the plaintext in the weak authenticity model.⁵ Nevertheless, we claim that introducing such an authenticity is meaningful to clarify what the security level can be realized by only the TI security.

Definition 11 (Weak Authenticity). $\mathcal{C}$ computes $\text{pp} \leftarrow \text{SetUp}(1^\lambda)$, $(\text{pk}_\sigma, \text{ek}_\sigma) \leftarrow \text{SKGen}(\text{pp}, \sigma)$, and $(\text{pk}_\rho, \text{dk}_\rho) \leftarrow \text{RKGGen}(\text{pp}, \rho)$, and gives $(\text{pp}, \text{pk}_\sigma, \text{pk}_\rho)$ to $\mathcal{A}$. $\mathcal{A}$ outputs (ct^*, M^*) where $M^* \notin \{\perp, \text{Uncertain}\}$. $\mathcal{C}$ computes $M \leftarrow \text{Dec}(\text{pp}, \text{dk}_\rho, \text{pk}_\sigma, \text{ct}^*)$. We say that PK-ME provides weak authenticity if for all PPT adversaries $\mathcal{A}$, the advantage

$$\text{Adv}_{\text{PK-ME}, \mathcal{A}}^{\text{wAuth}}(\lambda) = \Pr[M^* = M]$$

is negligible in the security parameter λ .

4 Proposed Generic Construction

In this section, we give the proposed generic construction of PK-ME from PAEKS. Before giving our construction, we revisited a generic construction of IBE from PEKS given by Boneh et al. [5] because our construction can be seen as an extension of the generic construction.

Generic construction of IBE from PEKS [5]. For an identity ID and a plaintext $M \in \{0, 1\}$, $ID||M$ is regarded as a keyword and is encrypted by the underlying PEKS scheme. The decryption

⁵For example, any two group elements can be regarded as a ciphertext of the ElGamal PKE scheme. That is, anyone can produce an ElGamal ciphertext without knowing the plaintext.

key consists of two trapdoors for two keywords $ID||0$ and $ID||1$. The IBE decryption algorithm outputs M if the test algorithm with the trapdoor of $ID||M$ outputs 1 where $M \in \{0, 1\}$. Due to the security of PEKS, no information of keyword (here $ID||M$) is revealed from PEKS ciphertext. Thus, no information of plaintext is revealed from IBE ciphertext. To simulate a decryption key query in (the IND-CPA security of) IBE, the reduction algorithm can employ the trapdoor oracle of the underlying PEKS scheme.

Why Uncertain is introduced in our construction? To decrypt a ciphertext via the search functionality of the underlying searchable encryption, we need to consider the following “uncertain” case where the test algorithm outputs 1 in both cases. Concretely, assume that the test algorithm outputs 1 with the trapdoor of $ID||0$, and also outputs 1 with the trapdoor of $ID||1$. Then, the decryption result is not uniquely determined. This uncertain case can be excluded if the underlying PEKS scheme provides consistency (though Boneh et al. did not explicitly mention it). This is the reason why we introduced “Uncertain” in the syntax. Since consistency is defined as a computational security manner in many cases,⁶ we define correctness of PK-ME (correctness 2) as a computational security manner.

Rhee and Lee [38] presented a generic construction of anonymous identity-based encryption (IBE) based on public-key encryption with keyword search (PEKS). They defined correctness in computational terms and asserted that the proposed anonymous IBE construction is correct if the underlying PEKS scheme is computationally consistent. However, the correctness of an IBE constructed from PEKS using Rhee and Lee’s generic construction cannot be guaranteed solely by consistency. This is because a consistent PEKS scheme lacking correctness can be easily devised, specifically, a scheme in which the test algorithm always outputs 0. Therefore, in this paper, we define the correctness of PK-ME separately, distinguishing between the portion ensured by the correctness of PAEKS and the portion ensured by the consistency of PAEKS.

Proposed Generic Construction. In our construction, a plaintext $M \in \{0, 1\}$ is regarded as a keyword, and is encrypted by the underlying PAEKS scheme. The decryption algorithm runs the test algorithm using trapdoors of keywords 0 and 1. Though we consider the case that the decryption result is Uncertain, this case can be excluded due to consistency of the underlying PAEKS scheme, and does not happen. Unlike the Boneh et al.’s construction [5], the proposed construction is not identity-based because the encryption algorithm takes a public key pk_{rcv} . We construct $\text{PK-ME} = (\text{Setup}, \text{SKGen}, \text{RKGen}, \text{Enc}, \text{Dec})$ from $\text{PAEKS} = (\text{PAEKS.Setup}, \text{PAEKS.KGen}_S, \text{PAEKS.KGen}_R, \text{PAEKS.Enc}, \text{Trapdoor}, \text{Test})$ as follows.

Setup(1^λ): Run $\text{pp}_{\text{PAEKS}} \leftarrow \text{PAEKS.Setup}(1^\lambda)$ and output $\text{pp} = \text{pp}_{\text{PAEKS}}$. Here, set $\mathcal{MSG} = \mathcal{KW} = \{0, 1\}$ and $\mathcal{IDX} = \{0, 1\}^\lambda$.

SKGen(pp, σ): Parse $\text{pp} = \text{pp}_{\text{PAEKS}}$. Run $(\text{PAEKS.pk}_\sigma, \text{PAEKS.sk}_\sigma) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$ and output $(\text{pk}_\sigma, \text{ek}_\sigma) = (\text{PAEKS.pk}_\sigma, \text{PAEKS.sk}_\sigma)$.

RKGen(pp, ρ): Parse $\text{pp} = \text{pp}_{\text{PAEKS}}$. Run $(\text{PAEKS.pk}_\rho, \text{PAEKS.sk}_\rho) \leftarrow \text{PAEKS.KGen}_R(\text{pp}_{\text{PAEKS}})$ and output $(\text{pk}_\rho, \text{sk}_\rho) = (\text{PAEKS.pk}_\rho, \text{PAEKS.sk}_\rho)$.

Enc($\text{pp}, \text{ek}_\sigma, \text{pk}_{\text{rcv}}, M$): Parse $\text{pp} = \text{pp}_{\text{PAEKS}}$, $\text{ek}_\sigma = \text{PAEKS.sk}_\sigma$, and $\text{pk}_{\text{rcv}} = \text{PAEKS.pk}_{\text{rcv}}$. Compute $\text{PAEKS.ct}_{\sigma, \text{rcv}, M} \leftarrow \text{PAEKS.Enc}(\text{PAEKS.pk}_\sigma, \text{PAEKS.pk}_{\text{rcv}}, \text{PAEKS.sk}_\sigma, M)$ and output $\text{ct} = \text{PAEKS.ct}_{\sigma, \text{rcv}, M}$.

⁶Some P(A)EKS schemes with statistical consistency have been proposed, e.g., [1, 36].

$\text{Dec}(\text{pp}, \text{dk}_\rho, \text{pk}_{\text{snd}}, \text{ct})$: Parse $\text{pp} = \text{pp}_{\text{PAEKS}}$, $\text{dk}_\rho = \text{PAEKS.sk}_\rho$, $\text{pk}_{\text{snd}} = \text{PAEKS.pk}_{\text{snd}}$, and $\text{ct} = \text{PAEKS.ct}_{\sigma, \text{rcv}, M}$. Compute $\text{td}_{\text{snd}, \rho, 0} \leftarrow \text{Trapdoor}(\text{PAEKS.pk}_{\text{snd}}, \text{PAEKS.pk}_\rho, \text{PAEKS.sk}_\rho, 0)$ and $\text{td}_{\text{snd}, \rho, 1} \leftarrow \text{Trapdoor}(\text{PAEKS.pk}_{\text{snd}}, \text{PAEKS.pk}_\rho, \text{PAEKS.sk}_\rho, 1)$.

- Output $M = 0$ if $\text{Test}(\text{PAEKS.ct}_{\sigma, \text{rcv}, M}, \text{td}_{\text{snd}, \rho, 0}) = 1 \wedge \text{Test}(\text{PAEKS.ct}_{\sigma, \text{rcv}, M}, \text{td}_{\text{snd}, \rho, 1}) = 0$.
- Output $M = 1$ if $\text{Test}(\text{PAEKS.ct}_{\sigma, \text{rcv}, M}, \text{td}_{\text{snd}, \rho, 0}) = 0 \wedge \text{Test}(\text{PAEKS.ct}_{\sigma, \text{rcv}, M}, \text{td}_{\text{snd}, \rho, 1}) = 1$.
- Output Uncertain if $\text{Test}(\text{PAEKS.ct}_{\sigma, \text{rcv}, M}, \text{td}_{\text{snd}, \rho, 0}) = 1 \wedge \text{Test}(\text{PAEKS.ct}_{\sigma, \text{rcv}, M}, \text{td}_{\text{snd}, \rho, 1}) = 1$.
- Output $\perp$ if $\text{Test}(\text{PAEKS.ct}_{\sigma, \text{rcv}, M}, \text{td}_{\text{snd}, \rho, 0}) = 0 \wedge \text{Test}(\text{PAEKS.ct}_{\sigma, \text{rcv}, M}, \text{td}_{\text{snd}, \rho, 1}) = 0$.

5 Security Analysis

Correctness. If the underlying PAEKS scheme is correct, then the proposed construction obviously provides correctness 1. We prove that the proposed construction provides correctness 2 if the underlying PAEKS is consistent. Intuitively, if the decryption result is Uncertain, then the test algorithm outputs 1 regardless of the keyword to be searched. This violates consistency.

Theorem 1. *If the underlying PAEKS scheme is consistent, then the proposed construction provides correctness 2.*

Proof. Let $\mathcal{C}$ be the challenger $\mathcal{A}$ be the adversary to correctness 2. We construct a reduction algorithm $\mathcal{B}$ that breaks consistency of PAEKS as follows. First, $\mathcal{C}$ computes $\text{pp}_{\text{PAEKS}} \leftarrow \text{PAEKS.SetUp}(1^\lambda)$, $(\text{PAEKS.pk}_S, \text{PAEKS.sk}_S) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$, $(\text{PAEKS.pk}_R, \text{PAEKS.sk}_R) \leftarrow \text{PAEKS.KGen}_R(\text{pp}_{\text{PAEKS}})$, and gives $(\text{pp}_{\text{PAEKS}}, \text{PAEKS.pk}_S$ and $\text{PAEKS.pk}_R)$ to $\mathcal{B}$. $\mathcal{B}$ sets $\text{pp} = \text{pp}_{\text{PAEKS}}$ and $(\text{pk}_\sigma, \text{pk}_\rho) = (\text{PAEKS.pk}_S, \text{PAEKS.pk}_R)$, and gives $(\text{pp}, \text{pk}_\sigma, \text{pk}_\rho)$ to $\mathcal{A}$. $\mathcal{A}$ outputs M . Since $\text{Dec}(\text{pp}, \text{dk}_\rho, \text{pk}_\sigma, \text{ct}) = \text{Uncertain}$ where $\text{ct} = \text{PAEKS.ct}_{\sigma, \text{rcv}, M}$, $\text{Test}(\text{PAEKS.ct}_{\sigma, \text{rcv}, M}, \text{td}_{\text{snd}, \rho, 0}) = 1 \wedge \text{Test}(\text{PAEKS.ct}_{\sigma, \text{rcv}, M}, \text{td}_{\text{snd}, \rho, 1}) = 1$ hold. $\mathcal{B}$ outputs M and breaks correctness 2. $\square$

Privacy. Next, we prove that the proposed construction provides privacy. The proof is relatively straightforward because no information of keyword is leaked from ciphertext if the underlying PAEKS scheme provides the full CI security. The underlying PAEKS scheme is also required to be key private because privacy guarantees that no information of (σ, rcv) is leaked from ciphertext.

Theorem 2. *If the underlying PAEKS scheme provides the full CI security with key privacy, then the proposed construction provides privacy.*

Proof. Let $\mathcal{C}$ be the challenger $\mathcal{A}$ be the adversary of privacy. We construct a reduction algorithm $\mathcal{B}$ that breaks the full CI security of PAEKS as follows. First, $\mathcal{C}$ computes $\text{pp}_{\text{PAEKS}} \leftarrow \text{PAEKS.SetUp}(1^\lambda)$, $(\text{PAEKS.pk}_{S_0}, \text{PAEKS.sk}_{S_0}) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$, $(\text{PAEKS.pk}_{S_1}, \text{PAEKS.sk}_{S_1}) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$, $(\text{PAEKS.pk}_{R_0}, \text{PAEKS.sk}_{R_0}) \leftarrow \text{PAEKS.KGen}_R(\text{pp}_{\text{PAEKS}})$, $(\text{PAEKS.pk}_{S_1}, \text{PAEKS.sk}_{S_1}) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$, and gives $(\text{PAEKS.pk}_{S_0}, \text{PAEKS.pk}_{S_1}, \text{PAEKS.pk}_{R_0}, \text{PAEKS.pk}_{R_1})$ to $\mathcal{B}$. $\mathcal{B}$ sets $\text{pk}_{\sigma_0} = \text{PAEKS.pk}_{S_0}$, $\text{pk}_{\sigma_1} = \text{PAEKS.pk}_{S_1}$, $\text{pk}_{\rho_0} = \text{PAEKS.pk}_{R_0}$, $\text{pk}_{\rho_1} = \text{PAEKS.pk}_{R_1}$, and gives $(\text{pk}_{\sigma_0}, \text{pk}_{\sigma_1}, \text{pk}_{\rho_0}, \text{pk}_{\rho_1})$ to $\mathcal{A}$.

When $\mathcal{B}$ receives an encryption query (i, j, M) from $\mathcal{A}$, $\mathcal{B}$ sends (i, j, M) to $\mathcal{C}$ as an encryption query where M is regarded as a keyword. $\mathcal{C}$ computes $\text{PAEKS.ct}_{S_i, R_j, M} \leftarrow \text{PAEKS.Enc}(\text{PAEKS.pk}_{S_i}, \text{PAEKS.pk}_{R_j}, \text{PAEKS.sk}_{S_i}, M)$ and returns $\text{PAEKS.ct}_{S_i, R_j, M}$ to $\mathcal{B}$. $\mathcal{B}$ sets $\text{ct} = \text{PAEKS.ct}_{S_i, R_j, M}$, and returns ct to $\mathcal{A}$.

In the challenge phase, $\mathcal{A}$ declares (M_0^*, M_1^*) . We set $(M_0^*, M_1^*) = (0, 1)$ without loss of generality. $\mathcal{B}$ sends (M_0^*, M_1^*) to $\mathcal{C}$ as the challenge keywords. $\mathcal{C}$ chooses $b \xleftarrow{\$} \{0, 1\}$, computes the challenge ciphertext $\text{PAEKS.ct}_{S_b, R_b, M_b^*}^* \leftarrow \text{PAEKS.Enc}(\text{PAEKS.pk}_{S_b}, \text{PAEKS.pk}_{R_b}, \text{PAEKS.sk}_{S_b}, M_b^*)$, and returns $\text{PAEKS.ct}_{S_b, R_b, M_b^*}^*$ to $\mathcal{B}$. $\mathcal{B}$ sets $\text{ct}^* = \text{PAEKS.ct}_{S_b, R_b, M_b^*}^*$ and sends ct^* to $\mathcal{A}$.

Finally, $\mathcal{A}$ outputs $b' \in \{0, 1\}$ to $\mathcal{B}$ and $\mathcal{B}$ outputs b' . Therefore, $\mathcal{A}$ breaks the full CI security with key privacy with the same advantage of $\mathcal{A}$. $\square$

Authenticity. Next, we prove that the proposed construction provides authenticity. Intuitively, the (full) TI security is not required to prove the following theorem since PK-ME does not consider information leakage from decryption keys. Counterintuitively, we reduce authenticity to the full TI security in our security proof. The intuition of the proof is explained as follows. According to the security definition, an adversary of authenticity outputs a decryptable ciphertext ct^* without possessing the corresponding encryption key. The reduction algorithm $\mathcal{B}$ tries to distinguish the challenge bit $b \in \{0, 1\}$ from the challenge trapdoor td_{S, R, kw_b^*} , where kw_b^* is either 0 or 1 in our construction. We focus on the fact that the challenge keywords are not allowed to be sent to the encryption oracle in the security definition of the (full) TI security, whereas ct^* is a ciphertext of a challenge keyword. Thus, it is expected that the test result of ct^* using td_{S, R, kw_b^*} reveals information of b . However, the test result does not help to break the TI security unless the decryption result of ct^* is explicitly known. That is, since ct^* is just required to be a decryptable ciphertext, the decryption result is not obvious (it is only guaranteed to be 0 or 1). If $\mathcal{B}$ knows that the decryption result of ct^* is 0 or 1, then $\mathcal{B}$ can recover b from the test result of ct^* using td_{S, R, kw_b^*} . To decrypt ct^* , $\mathcal{B}$ sends the challenge keywords to $\mathcal{C}$ as trapdoor queries, and decrypts ct^* using the trapdoors. This is the reason why the underlying PAEKS scheme needs to be full TI secure.

Theorem 3. *If the underlying PAEKS scheme provides the full TI security, then the proposed construction provides authenticity.*

Proof. Let $\mathcal{C}$ be the challenger $\mathcal{A}$ be the adversary of authenticity. We construct a reduction algorithm $\mathcal{B}$ that breaks the full TI security of PAEKS as follows. First, $\mathcal{C}$ computes $\text{pp}_{\text{PAEKS}} \leftarrow \text{PAEKS.SetUp}(1^\lambda)$, $(\text{PAEKS.pk}_S, \text{PAEKS.sk}_S) \leftarrow \text{PAEKS.KGen}_S(\text{pp}_{\text{PAEKS}})$, and $(\text{PAEKS.pk}_R, \text{PAEKS.sk}_R) \leftarrow \text{PAEKS.KGen}_R(\text{pp}_{\text{PAEKS}})$, and sends $(\text{pp}_{\text{PAEKS}}, \text{PAEKS.pk}_S, \text{PAEKS.pk}_R)$ to $\mathcal{B}$. $\mathcal{B}$ declares the challenge keywords $(kw_0^*, kw_1^*) = (0, 1)$, and sends it to $\mathcal{C}$. $\mathcal{C}$ chooses $b \xleftarrow{\$} \{0, 1\}$, computes the challenge trapdoor $\text{td}_{S, R, kw_b^*} \leftarrow \text{Trapdoor}(\text{PAEKS.pk}_S, \text{PAEKS.pk}_R, \text{PAEKS.sk}_R, kw_b^*)$, and sends td_{S, R, kw_b^*} to $\mathcal{B}$.

$\mathcal{B}$ sends $kw_0^* = 0$ to $\mathcal{C}$ as a trapdoor query. $\mathcal{C}$ computes $\text{td}_{S, R, kw_0^*} \leftarrow \text{Trapdoor}(\text{PAEKS.pk}_S, \text{PAEKS.pk}_R, \text{PAEKS.sk}_R, kw_0^*)$, and sends td_{S, R, kw_0^*} to $\mathcal{B}$. $\mathcal{B}$ also sends $kw_1^* = 1$ to $\mathcal{C}$ as a trapdoor query. $\mathcal{C}$ computes $\text{td}_{S, R, kw_1^*} \leftarrow \text{Trapdoor}(\text{PAEKS.pk}_S, \text{PAEKS.pk}_R, \text{PAEKS.sk}_R, kw_1^*)$, and sends td_{S, R, kw_1^*} to $\mathcal{B}$.

$\mathcal{B}$ sets $\text{pk}_\sigma = \text{PAEKS.pk}_S$, $\text{pk}_\rho = \text{PAEKS.pk}_R$, and $\text{pp} = \text{pp}_{\text{PAEKS}}$, and sends $(\text{pp}, \text{pk}_\sigma, \text{pk}_\rho)$ to $\mathcal{A}$. $\mathcal{A}$ outputs ct^* . Let M^* be the decryption result of ct^* . Since ct^* is a decryptable ciphertext and thus $M^* \in \{0, 1\}$ and $M^* \notin \{\text{Uncertain}, \perp\}$ hold, we do not have to consider the cases $\text{Test}(\text{ct}^*, \text{td}_{S, R, kw_0^*}) = 1 \wedge \text{Test}(\text{ct}^*, \text{td}_{S, R, kw_1^*}) = 1$ and $\text{Test}(\text{ct}^*, \text{td}_{S, R, kw_0^*}) = 0 \wedge \text{Test}(\text{ct}^*, \text{td}_{S, R, kw_1^*}) = 0$. $\mathcal{B}$ recovers b as follows.

- If $\text{Test}(\text{ct}^*, \text{td}_{S, R, kw_0^*}) = 1 \wedge \text{Test}(\text{ct}^*, \text{td}_{S, R, kw_1^*}) = 0$, then $M^* = 0$. If $\text{Test}(\text{ct}^*, \text{td}_{S, R, kw_b^*}) = 0$, then $b = 1$. If $\text{Test}(\text{ct}^*, \text{td}_{S, R, kw_b^*}) = 1$, then $b = 0$.
- If $\text{Test}(\text{ct}^*, \text{td}_{S, R, kw_0^*}) = 0 \wedge \text{Test}(\text{ct}^*, \text{td}_{S, R, kw_1^*}) = 1$, then $M^* = 1$. If $\text{Test}(\text{ct}^*, \text{td}_{S, R, kw_b^*}) = 0$, then $b = 0$. If $\text{Test}(\text{ct}^*, \text{td}_{S, R, kw_b^*}) = 1$, then $b = 1$.

Table 1: PAEKS for Instantiating the Proposed Construction

PAEKS (via the Li-Boyen Generic Construction [31])	Authenticity of PK-ME	Complexity Assumption	Standard Model
EPE [39]+NIKE [18]	Full	MDDH	Yes
EPE [31]+NIKE [20, 27]	Full	MLWE NTRU	No
EPE [31] +NIKE [17, 20]	Full	MLWE NTRU	Yes
HMAC+NIKE [18]	Weak	DDH	Yes
HMAC+NIKE [8]	Weak	CDH	Yes
HMAC+NIKE [26]	Weak	DBDH	Yes
HMAC+NIKE [26]	Weak	Factoring	No
HMAC+NIKE [9]	Weak	CSIDH	Yes
HMAC+NIKE [27]	Weak	MLWE	No
HMAC+NIKE [27]	Weak	MLWE	Yes

$\mathcal{B}$ outputs b , and breaks the full TI security with the same advantage of $\mathcal{A}$. $\square$

Obviously, the following theorem holds. The proof is the same as that of Theorem 3 except that an adversary declares (ct^*, M^*) where $M^* \in \{0, 1\}$. Thus, the reduction algorithm does not have to issue a trapdoor query for the challenge keywords.

Theorem 4. *If the underlying PAEKS scheme provides the TI security, then the proposed construction provides weak authenticity.*

6 Instantiations

Here, we summarize the underlying PAEKS schemes that can be employed for instantiating the proposed generic construction in Table 1. We describe EPE and NIKE for instantiating the Li-Boyen generic construction of PAEKS [31]. If the underlying PAEKS scheme provides the full TI security (resp. the TI security), then the PK-ME scheme instantiated by the PAEKS scheme provides authenticity (resp. weak authenticity). We denote Full if the PK-ME scheme provides authenticity. Here, the lattice-based EPE is given by Li-Boyen [31]. Shinoki et al. [39] pointed out a flaw of the Li-Boyne pairing EPE and proposed a MDDH (Matrix Decisional Diffie-Hellman) based EPE. Thus, we employ the Shinoki et al.’s EPE in the table. Here, DBDH stands for Decisional Bilinear Diffie-Hellman, MLWE stands for Module Learning with Errors, and CSIDH stands for Commutative Supersingular isogeny Diffie-Hellman. Via the proposed construction with these PAEKS schemes, we can obtain PK-ME schemes with several complexity assumptions.

7 Discussion

In this section, we compare PK-ME with IB-ME and explore its advantages and disadvantages.

7.1 Advantages of PK-ME

Security against the KGC: In IBE including IB-ME, the KGC generates secret keys for each identity. Thus, the KGC potentially can decrypt all ciphertexts. Though IB-ME secure

against the KGC has been proposed in [13], the scheme requires additional authority. This key escrow problem does not happen in the public key encryption setting including PK-ME at the expense of introducing public key infrastructure (PKI) (see below).

Efficient Instantiation: Boneh et al. demonstrated that no blackbox construction of IBE exists from PKE [6]. Thus, basically, PKE is efficiently instantiated compared to IBE. Though an IBE construction from the computational Diffie-Hellman assumption has been proposed by Döttling and Garg [19], it requires garbled circuits and is much less efficient compared to those of pairing-based IBE schemes. Since PK-ME is a public key encryption variant, PK-ME is not affected by the impossibility result. For instantiating a PK-ME scheme with authenticity, the current security proof requires full TI security, and current PAEKS schemes with full TI security were realized by IBE-related techniques. For instantiating a PK-ME scheme with weak authenticity, PK-ME can be instantiated without employing IBE-related techniques.

7.2 Disadvantages of PK-ME

PKI: In exchange for the absence of the key escrow problem, it is necessary to assume a PKI system. In this paper, it is implicitly assumed that the PKI issues a certificate for each public key. Issuance of public key certificates by PKI is a cost and is a disadvantage of PK-ME.

Impossibility of Enhanced Privacy: Francati et al. [24] introduced a stronger flavor of privacy which they call enhanced privacy. It captures privacy in IB-ME according to every possible mismatch condition for the receiver where an adversary holds a decryption key but does not know the sender's identities. Enhanced privacy implicitly requires an exponentially large number of identities (public keys). In the security model, two challenge sender's identities σ_0 and σ_1 are chosen from the identity space by the challenger. Here, the space is required to have a non-trivial amount of min-entropy. Obviously, the adversary wins if the size of the identity space is polynomially bounded. Due to this observation, PK-ME has no way to provide enhanced privacy since the number of public keys is bounded by a polynomial of the security parameter. Nonetheless, we would like to claim that our definitions, privacy and authenticity, capture fundamental security notions of PK-ME similar to those of IB-ME defined by Ateniese et al. [3,4]. Simultaneously, we do not deny any possibility to strengthen the privacy or authenticity of PK-ME. Considering such an enhanced security of PK-ME is left as future work.

7.3 Applications

Next, we explore whether PK-ME can be used for realizing applications of IB-ME. Ateniese et al. [3,4] considered an application of IB-ME where “*an FBI agent may want to communicate only with CIA agents, and if this is not the case, the communication should drop without revealing membership information and why the communication failed.*”. The sender can specify that the message is intended for an FBI agent during encryption, and the receiver can also specify that he/she wants to read messages only if they come from CIA agents during decryption. In this example, the KGC issues keys after confirming the attributes. Ateniese et al. explicitly mentioned that “*The sender's attributes are certified by the authority, so that no malicious sender can forge a valid ciphertext which embeds fake attributes.*”. The receiver's attributes are also certified by the KGC. That is, we can employ PK-ME for this application if we assume that PKI confirms the sender's and receiver's attributes and issues certificates for their public keys.

Ateniese et al. constructed an anonymous bulletin board using IB-ME and Tor. Senders post encrypted messages to a bulletin board, hosted by a Tor hidden service. As well as the encryption key corresponding to their own identity, senders specify a target identity string that acts as the receiver's access policy. Receivers also specify the expected identity of the sender. Regarding the key generation by the KGC, Ateniese et al. mentioned that the key generation service could be implemented as another Tor hidden service that automatically converts email addresses or phone numbers into keys. Ateniese et al. also mentioned the key escrow problem where “*We stress that ME is susceptible to key escrow attacks. For this reason, the key generation service must be trusted. Promising approaches to mitigate (or solve) key escrow attacks in ME (not considered in this paper) are the ones that have been developed for identity-based and attribute-based cryptosystems, e.g., multi-authorities or the registration-based setting.*”. Let assume that PKI issues a certificate on, e.g., a public key||identity, where the identity (e.g., email addresses or phone numbers as mentioned above) has the same role of those in the IB-ME based anonymous bulletin board system. Then, we can construct a key escrow free anonymous bulletin board system using PK-ME and Tor at the expense of the cost of the PKI system. To sum up, the public key setting does not affect the above applications of IB-ME.

8 Conclusion

In this paper, we focus on a similarity of matchmaking encryption and PAEKS. We formalize PK-ME and propose a generic construction of PK-ME from PAEKS. Our security proof of authenticity requires that the underlying PAEKS scheme is to be full TI secure. Thus, we can instantiate the proposed generic construction from pairing or lattice as mentioned before. The proposed construction considers 1-bit plaintext. It is expected that the plaintext space can be amplified by simply arranging ciphertexts of 1-bit plaintext. Exploring more an efficient way for plaintext amplification is left as future work of this paper.

Our work indicates some interesting open problems in both PK-ME and PAEKS contexts. For privacy, we weaken the security level by defining the encryption oracle. If an adversary is allowed to obtain sender's secret keys in the full CI security model, then we can enhance the security level of privacy by giving the challenge encryption key to the adversary. However, the Li-Boyen generic construction of PAEKS seems insecure when an adversary has the sender's secret key because the adversary can produce NIKE keys (we emphasize that this is not an attack against the Li-Boyen construction). Considering a secret key exposure attack in PAEKS itself seems interesting topic to enhance the security level of full CI in PAEKS, and it also enhances the security level of privacy in PK-ME via the proposed generic construction.

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References

- [1] Michel Abdalla, Mihir Bellare, Dario Catalano, Eike Kiltz, Tadayoshi Kohno, Tanja Lange, John Malone-Lee, Gregory Neven, Pascal Paillier, and Haixia Shi. Searchable encryption revisited: Consistency properties, relation to anonymous IBE, and extensions. *Journal of Cryptology*, 21(3):350–391, 2008.
- [2] Joël Alwen, Dominik Hartmann, Eike Kiltz, Marta Mularczyk, and Peter Schwabe. Post-quantum multi-recipient public key encryption. In *ACM CCS*, pages 1108–1122, 2023.

- [3] Giuseppe Ateniese, Danilo Francati, David Nuñez, and Daniele Venturi. Match me if you can: Matchmaking encryption and its applications. In *CRYPTO*, pages 701–731, 2019.
- [4] Giuseppe Ateniese, Danilo Francati, David Nuñez, and Daniele Venturi. Match me if you can: Matchmaking encryption and its applications. *Journal of Cryptology*, 34(3):16, 2021.
- [5] Dan Boneh, Giovanni Di Crescenzo, Rafail Ostrovsky, and Giuseppe Persiano. Public key encryption with keyword search. In *EUROCRYPT*, pages 506–522, 2004.
- [6] Dan Boneh, Periklis A. Papakonstantinou, Charles Rackoff, Yevgeniy Vahlis, and Brent Waters. On the impossibility of basing identity based encryption on trapdoor permutations. In *IEEE FOCS*, pages 283–292, 2008.
- [7] Xavier Boyen and Qinyi Li. Identity-based matchmaking encryption with enhanced privacy - A generic construction with practical instantiations. In *ESORICS*, pages 425–445, 2023.
- [8] David Cash, Eike Kiltz, and Victor Shoup. The twin Diffie-Hellman problem and applications. In *EUROCRYPT*, pages 127–145, 2008.
- [9] Wouter Castryck, Tanja Lange, Chloe Martindale, Lorenz Panny, and Joost Renes. CSIDH: an efficient post-quantum commutative group action. In *ASIACRYPT*, pages 395–427, 2018.
- [10] Biwen Chen, Tao Xiang, Mimi Ma, Debiao He, and Xiaofeng Liao. CL-ME: efficient certificateless matchmaking encryption for internet of things. *IEEE Internet Things J.*, 8(19):15010–15023, 2021.
- [11] Jie Chen, Yu Li, Jinming Wen, and Jian Weng. Identity-based matchmaking encryption from standard assumptions. In *ASIACRYPT*, pages 394–422, 2022.
- [12] Leixiao Cheng, Jing Qin, Feng Feng, and Fei Meng. Security-enhanced public-key authenticated searchable encryption. *Information Sciences*, 647:119454, 2023.
- [13] Sohto Chiku, Keisuke Hara, and Junji Shikata. Identity-based matchmaking encryption secure against key generation center. In *ProvSec*, pages 251–273, 2023.
- [14] Sohto Chiku, Keitaro Hashimoto, Keisuke Hara, and Junji Shikata. Identity-based matchmaking encryption, revisited: Strong security and practical constructions from standard classical and post-quantum assumptions. *IACR Cryptology ePrint Archive*, page 1435, 2023.
- [15] Sherman S. M. Chow. Removing escrow from identity-based encryption. In *Public Key Cryptography*, pages 256–276, 2009.
- [16] Qiaohan Chu, Anmin Fu, Haifeng Qian, and Jie Chen. Inner-product matchmaking encryption: Bilateral access control and beyond equality. *IET Information Security*, 2023:1–17, 2023.
- [17] Bor de Kock. A non-interactive key exchange based on ring-learning with errors, 2018. PhD thesis, Master ’ s thesis. Master ’ s thesis, Eindhoven University of Technology.
- [18] Whitfield Diffie and Martin E. Hellman. New directions in cryptography. *IEEE Transactions on Information Theory*, 22(6):644–654, 1976.
- [19] Nico Döttling and Sanjam Garg. Identity-based encryption from the Diffie-Hellman assumption. In *CRYPTO*, pages 537–569, 2017.

- [20] Léo Ducas, Vadim Lyubashevsky, and Thomas Prest. Efficient identity-based encryption over NTRU lattices. In *ASIACRYPT*, pages 22–41, 2014.
- [21] Keita Emura. Generic construction of public-key authenticated encryption with keyword search revisited: Stronger security and efficient construction. In *ACM APKC*, pages 39–49, 2022.
- [22] Keita Emura, Shuichi Katsumata, and Yohei Watanabe. Identity-based encryption with security against the KGC: A formal model and its instantiation from lattices. In *ESORICS*, pages 113–133, 2019.
- [23] Keita Emura, Shuichi Katsumata, and Yohei Watanabe. Identity-based encryption with security against the KGC: A formal model and its instantiations. *Theoretical Computer Science*, 900:97–119, 2022.
- [24] Danilo Francati, Alessio Guidi, Luigi Russo, and Daniele Venturi. Identity-based matchmaking encryption without random oracles. In *INDOCRYPT*, pages 415–435, 2021.
- [25] Danilo Francati, Valeria Huang, and Daniele Venturi. Registered matchmaking encryption. In *Applied Cryptography and Network Security*, 2025, to appear.
- [26] Eduarda S. V. Freire, Dennis Hofheinz, Eike Kiltz, and Kenneth G. Paterson. Non-interactive key exchange. In *Public-Key Cryptography*, pages 254–271, 2013.
- [27] Phillip Gajland, Bor de Kock, Miguel Quaresma, Giulio Malavolta, and Peter Schwabe. SWOOSH: efficient lattice-based non-interactive key exchange. In *USENIX Security*, 2024.
- [28] Martha Norberg Hovd and Martijn Stam. Vetted encryption. In *INDOCRYPT*, pages 488–507, 2020.
- [29] Qiong Huang and Hongbo Li. An efficient public-key searchable encryption scheme secure against inside keyword guessing attacks. *Information Sciences*, 403:1–14, 2017.
- [30] Zhe Jiang, Xiwen Wang, Kai Zhang, Junqing Gong, Jie Chen, and Haifeng Qian. Revocable identity-based matchmaking encryption in the standard model. *IET Information Security*, 17(4):567–581, 2023.
- [31] Qinyi Li and Xavier Boyen. Public-key authenticated encryption with keyword search made easy. *IACR Communications in Cryptology*, 1(2):16, 2024.
- [32] Shen Lin, Yu Li, and Jie Chen. CCA-secure identity-based matchmaking encryption from standard assumptions. In *Inscrypt*, pages 253–273, 2023.
- [33] Yunhao Ling, Guang Zhang, Jie Chen, and Haifeng Qian. Efficient one-to-one sharing: Public key matchmaking encryption. *Journal of Systems Architecture*, 167:103492, 2025.
- [34] John Malone-Lee. Identity-based signcryption. *IACR Cryptol. ePrint Arch.*, page 98, 2002.
- [35] Alberto Maria Mongardini, Daniele Friolo, and Giuseppe Ateniese. Stealth and beyond: Attribute-driven accountability in Bitcoin transactions. IACR Cryptology ePrint Archive, Paper 2024/1789, 2024.
- [36] Sayantan Mukherjee. Statistically consistent broadcast authenticated encryption with keyword search - adaptive security from standard assumptions. In *ACISP*, pages 523–552, 2023.

- [37] Kenneth G. Paterson and Sriramkrishnan Srinivasan. On the relations between non-interactive key distribution, identity-based encryption and trapdoor discrete log groups. *Designs, Codes and Cryptography*, 52(2):219–241, 2009.
- [38] Hyun Sook Rhee and Dong Hoon Lee. Anonymous IBE from PEKS: A generic construction. In *WISA*, pages 105–118, 2021.
- [39] Hirotomo Shinoki, Hisayoshi Sato, and Masayuki Yoshino. Fully secure searchable encryption from PRFs, pairings, and lattices. In *Applied Cryptography and Network Security, Part I*, pages 62–88, 2025.
- [40] Xiwen Wang, Kai Zhang, Jinguo Li, Mi Wen, Shengmin Xu, and Jianting Ning. Non-interactive boolean searchable asymmetric encryption with bilateral access control. *The Computer Journal*, 67(1):179–194, 2024.
- [41] Yuejun Wang, Baocang Wang, Qiqi Lai, and Yu Zhan. Identity-based matchmaking encryption with stronger security and instantiation on lattices. *Theoretical Computer Science*, 1029:115048, 2025.
- [42] Ningbin Yang, Chunming Tang, and Debiao He. A lightweight certificateless multi-user matchmaking encryption for mobile devices: Enhancing security and performance. *IEEE Transactions on Information Forensics and Security*, 19:251–264, 2024.
- [43] Takeshi Yoshida and Keita Emura. On the relations between matchmaking public key encryption and public key authenticated encryption with keyword search. In *CANS*, pages 272–291, 2025.